The Foundations: Logic and Proofs

Chapter 1, Part II: Predicate Logic

With Question/Answer Animations

Predicates and Quantifiers

Section 1.4

Section Summary

- Predicates
- Variables
- Quantifiers
  - Universal Quantifier
  - Existential Quantifier
- Negating Quantifiers
  - De Morgan's Laws for Quantifiers
- Translating English to Logic
- Logic Programming (*optional*)

Propositional Logic Not Enough

- If we have:

  "All men are mortal."

  "Socrates is a man."
- Does it follow that "Socrates is mortal?"
- Can't be represented in propositional logic. Need a language that talks about objects, their properties, and their relations.
- Later we'll see how to draw inferences.

Introducing Predicate Logic

- Predicate logic uses the following new features:
  - Variables: x, y, z
  - Predicates: $P(x), M(x)$

    متغيرات
    و فنكشن
  - Quantifiers (*to be covered in a few slides*):
- *Propositional functions* are a generalization of propositions.
  - They contain variables and a predicate, e.g., $P(x)$
  - Variables can be replaced by elements from their *domain*.

Propositional Functions

- Propositional functions become propositions (and have truth values) when their variables are each replaced by a value from the *domain* (or *bound* by a quantifier, as we will see later).
- The statement *P(x)* is said to be the value of the propositional function *P* at *x*.
- For example, let *P(x)* denote "$x > 0$" and the domain be the integers. Then:

  P(-3) is false. $-3 > 0$ X

  P(0) is false. $0 > 0$ X

  P(3) is true. $3 > 0$ ✓
- Often the domain is denoted by *U*. So in this example *U* is the integers.

- The statement *P(x)* is said to be the value of the propositional function *P* at *x*.
- **Example**, let *P(x)* denote "$x > 0$" and the domain be the integers. Then:

  P(-3): -3>0 the truth value is false.

  P(0) : 0>0 the truth value is false.

  P(3) : 3>0 the truth value is true.
- Often the domain is denoted by ***U***. So in this example ***U*** is the integers.

Examples of Propositional Functions

- Let "$x + y = z$" be denoted by $R(x, y, z)$ and U (for all three variables) be the integers. Find these truth values:

  x y z

  R(2,-1,5) $2+(-1)=5$ $= 1$ x False

  Solution: F

  R(3,4,7) $3+4=7$ ✓ True

  Solution: T

  $R(x, 3, z)$

  Solution: Not a Proposition إذا الجملة تحتوي متغيران بس قيم

- Now let "$x - y = z$" be denoted by $Q(x, y, z)$, with U as the integers. Find these truth values:

  Q(2,-1,3)

  Solution: T

  Q(3,4,7) $3-4=-1$

  Solution: F

  $Q(x, 3, z)$

  Solution: Not a Proposition

Compound Expressions

- Connectives from propositional logic carry over to predicate logic.
- If *P(x)* denotes "*x* > 0," find these truth values:

  P(3) ∨ P(-1) **Solution**: T — $3>0 \lor -1>0 \longrightarrow$ True or false = T

  P(3) ∧ P(-1) **Solution**: F — $3>0 \land -1>0 \longrightarrow$ True AND false = F

  الاول T
  الثاني f
  f =

  P(3) → P(-1) **Solution**: F — $3>0 \rightarrow -1>0 \longrightarrow$ True implies false = F

  [اذا اطلع T , f]

  P(3) → ¬P(-1) **Solution**: T — $3>0 \rightarrow -1 \not> 0 \longrightarrow$ True implies True = T

  لا عشان نفي نعكس الاشارة

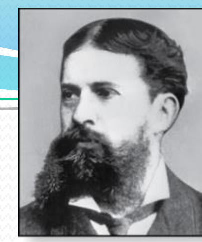
  عواقب هذا يصير T

- Expressions with variables are not propositions and therefore do not have truth values. For example,

  P(3) ∧ P(*y*)

  P(*x*) → P(*y*)

- When used with quantifiers (to be introduced next), these expressions (propositional functions) become propositions.



Charles Peirce (1839-1914)

Quantifiers

- We need *quantifiers* to express the meaning of English words including *all* and *some*:
  - "All men are Mortal."
  - "Some cats do not have fur."
- The two most important quantifiers are:
  - *Universal Quantifier*, "For all," symbol: $\forall$ All / every
  - *Existential Quantifier*, "There exists," symbol: $\exists$ exist / sym [يوجد]
- We write as in $\forall x\, P(x)$ and $\exists x\, P(x)$.
- $\forall x\, P(x)$ asserts $P(x)$ is true for <u>every</u> x in the *domain*.
- $\exists x\, P(x)$ asserts $P(x)$ is true for <u>some</u> x in the *domain*.
- The quantifiers are said to bind the variable x in these expressions.

Universal Quantifier

- $\forall x\, P(x)$ is read as "For all x, P(x)" or "For every x, P(x)"

Examples:

مجموعة الأرقام سالبة وموجبة مالانهاية

1) If $P(x)$ denotes "$x > 0$" and U is the integers, then $\forall x\, P(x)$ is false.

عشان يكون True لازم كل الأعداد اللي أعوضها تكون True

أعداد موجبة

2) If $P(x)$ denotes "$x > 0$" and U is the positive integers, then $\forall x\, P(x)$ is true.

3) If $P(x)$ denotes "x is even" and U is the integers, then $\forall x\, P(x)$ is false.

عدد زوجي

يوجد

Existential Quantifier

- $\exists x\, P(x)$ is read as "For some x, P(x)", or as "There is an x such that P(x)," or "For at least one x, P(x)."

Examples:

1. If $P(x)$ denotes "$x > 0$" and U is the integers, then $\exists x\, P(x)$ is true. It is also true if U is the positive integers.

1<0 ✗
2<0 ✗
3<0 ✗

2. If $P(x)$ denotes "$x < 0$" and U is the positive integers, then $\exists x\, P(x)$ is false.

[يوجد لو عدد واحد ينطبق الشرط يكون True]

3. If $P(x)$ denotes "x is even" and U is the integers, then $\exists x\, P(x)$ is true.

Ex

Properties of Quantifiers

- The truth value of $\exists x\, P(x)$ and $\forall x\, P(x)$ depend on both the propositional function $P(x)$ and on the domain U.
- **Examples**:
  1. If U is the positive integers and $P(x)$ is the statement "$x < 2$", then $\exists x\, P(x)$ is true, but $\forall x\, P(x)$ is false.
  2. If U is the negative integers and $P(x)$ is the statement "$x < 2$", then both $\exists x\, P(x)$ and $\forall x\, P(x)$ are true.
  3. If U consists of 3, 4, and 5, and $P(x)$ is the statement "$x > 2$", then both $\exists x\, P(x)$ and $\forall x\, P(x)$ are true. But if $P(x)$ is the statement "$x < 2$", then both $\exists x\, P(x)$ and $\forall x\, P(x)$ are false.

Precedence of Quantifiers

- The quantifiers $\forall$ and $\exists$ have higher precedence than all the logical operators.
- For example, $\forall x\, P(x) \lor Q(x)$ means $(\forall x\, P(x)) \lor Q(x)$
- $\forall x\, (P(x) \lor Q(x))$ means something different.
- Unfortunately, often people write $\forall x\, P(x) \lor Q(x)$ when they mean $\forall x\, (P(x) \lor Q(x))$.

Translating from English to Logic

Example 1: Translate the following sentence into predicate logic: "Every student in this class has taken a course in Java."

Solution:

First decide on the domain U.

Solution 1: If U is all students in this class, define a propositional function J(x) denoting "x has taken a course in Java" and translate as $\forall x\, J(x)$. كل

Solution 2: But if U is all people, also define a propositional function S(x) denoting "x is a student in this class"

"For every person x, if person x is a student in this class then x taken a course in Java."

طالب

and translate as $\forall x\,(S(x) \rightarrow J(x))$. إذا تحقق الشرط سوف يدرس

Then

$\forall x\,(S(x) \land J(x))$ is not correct. What does it mean?

غالبًا All تجي معها $(\rightarrow)$

Translating from English to Logic

Example 2: Translate the following sentence into predicate logic: "Some student in this class has taken a course in Java."

Solution:

First decide on the domain U.

Solution 1: If U is all students in this class, translate as

شخص Java

$\exists x\, J(x)$

يوجد

Solution 1: But if U is all people, then translate as

- "There is a person x having the properties that x is a student in this class and x has taken a course in Java."

AND

and translate as $\exists x\, (S(x) \wedge J(x))$

$\exists x\, (S(x) \rightarrow J(x))$ is not correct. What does it mean?

AND [illegible] some [illegible]

Returning to the Socrates Example

- Introduce the propositional functions *Man(x)* denoting "*x* is a man" and *Mortal(x)* denoting "*x* is mortal." Specify the domain as all people.
- The two premises are: $\forall x Man(x) \rightarrow Mortal(x)$

  $Man(Socrates)$
- The conclusion is: $Mortal(Socrates)$
- Later we will show how to prove that the conclusion follows from the premises.

النفي

Negating Quantified Expressions

- Consider $\forall x\, J(x)$ اسود بدون نفي واضيف النفي

  "Every student in your class has taken a course in Java."

  Here $J(x)$ is "x has taken a course in Java" and the domain is students in your class.

- Negating the original statement gives "It is not the case that every student in your class has taken Java." This implies that "There is a student in your class who has not taken Java."

  ∃ يوجد

  Symbolically $\neg\forall x\, J(x)$ and $\exists x\, \neg J(x)$ are equivalent

[Not every taken java] النفي برا [يوجد بعض لم يأخذ جافا]

Negating Quantified Expressions (*continued*)

انفي العبارة هذه

- Now Consider $\exists x\, J(x)$

$\exists$ "There is a student in this class who has taken a course in Java."

Where $J(x)$ is "x has taken a course in Java."

1

- Negating the original statement gives "It is not the case that there is a student in this class who has taken Java." This implies that "Every student in this class has not taken Java" 2

$\forall$

Symbolically $\neg\exists x\, J(x)$ and $\forall x\, \neg J(x)$ are $\boxed{\text{equivalent}}$

[لايوجد طلاب يدرسو جافا] [كل الطلاب لايدرسون جافا]

De Morgan's Laws for Quantifiers

- The rules for negating quantifiers are:

TABLE 2 De Morgan's Laws for Quantifiers.

Negation	*Equivalent Statement*	*When Is Negation True?*	*When False?*
$\neg\exists x P(x)$	$\forall x \neg P(x)$	For every x, $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg\forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x.

- The reasoning in the table shows that:

$$\neg\forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg\exists x P(x) \equiv \forall x \neg P(x)$$

- These are important. You will use these.

Translation from English to Logic

- **Example**: Express the statement "Every student in this class has studied calculus" using predicates and quantifiers.
- **Solution**: *First*, we rewrite the statement so that we can clearly identify the appropriate quantifiers to use. Doing so, we obtain:

  "For every student in this class, that student has studied calculus."

  Next, we introduce a variable x so that our statement becomes:

  "For every student x in this class, x has studied calculus.

  Now we introduce $C(x)$ denote "x has studied calculus."
- If the domain for x consists of the students in the class then $\forall x\, C(x)$ translate our statement, If we change the domain to consist of all people, we will need to express our statement as "For every person x, if person x is a student in this class then x has studied calculus."
- If S(x) represents the statement that person x is in this class, we see that our statement can be expressed as $\forall x(S(x) \rightarrow C(x))$.

Translation from English to Logic

- **Examples**: Express the statements "Some student in this class has visited Mexico" and "Every student in this class has visited either Canada or Mexico" using predicates and quantifiers.

1. "Some student in this class has visited Mexico."

   Solution: Let $M(x)$ denote "x has visited Mexico" and $S(x)$ denote "x is a student in this class," and U be all people.

   $$\exists x \ (S(x) \land M(x))$$

2. "Every student in this class has visited Canada or Mexico."

   Solution: Add $C(x)$ denoting "x has visited Canada."

   $$\forall x \ (S(x) \rightarrow (M(x) \lor C(x)))$$

System Specification Example

- Predicate logic is used for specifying properties that systems must satisfy.
- For example, translate into predicate logic:
  - "Every mail message larger than one megabyte will be compressed."
  - "If a user is active, at least one network link will be available."
- Decide on predicates and domains for the variables:
  - Let $L(m, y)$ be "Mail message m is larger than y megabytes."
  - Let $C(m)$ denote "Mail message m will be compressed."
  - Let $A(u)$ represent "User u is active."
  - Let $S(n, x)$ represent "Network link n is state x.
- Now we have: $\forall m(L(m,1) \rightarrow C(m))$

$$\exists u\, A(u) \rightarrow \exists n\, S(n, available)$$